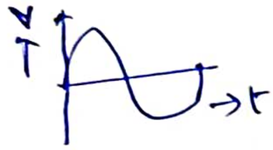
2 (ii) Oscillators

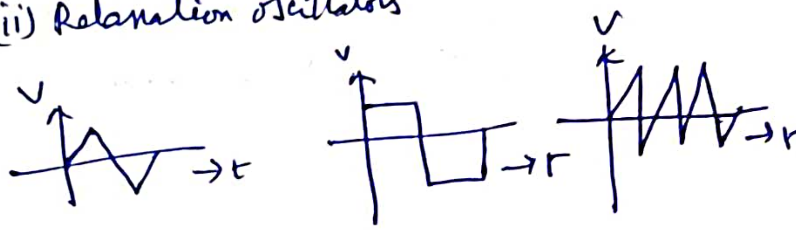
Any circuit which is used to generate an ac Voltage with out ac i/p signal is called an oscillator.

To generate the a c Voltage, the circuit is supplied with energy from a dc source.

Classification of oscillators :- I :- According to waveforms generated

(i) Sinusoidal oscillators (ii) Relaxation oscillators





II According to fundamental mechanisms involved.

(a) Negative resistance oscillators

(b) Feedback oscillators

Negative Resistance oscillator :- uses negative resistance of the amplifying device to neutralize the +ve resistance of the oscillator.

Feedback oscillator :- uses +ve feedback in the feedback amplifier to satisfy the Barkhausen conditions.

III According to frequency generated :-

(a) Audio frequency oscillator : upto 20 kHz (b) RFO : 20 kHz to 30 MHz

(c) VHF oscillators : 30 MHz to 300 MHz (d) UHF oscillators : 300 MHz to 3 GHz

(e) Microwave frequency oscillator : above 3 GHz.

IV According to type of circuit used, Sinusoidal oscillators may be classified as

(a) LC tuned oscillator

(b) RC phase shift oscillator

Conditions for oscillation:- (Barkhausen criterion)

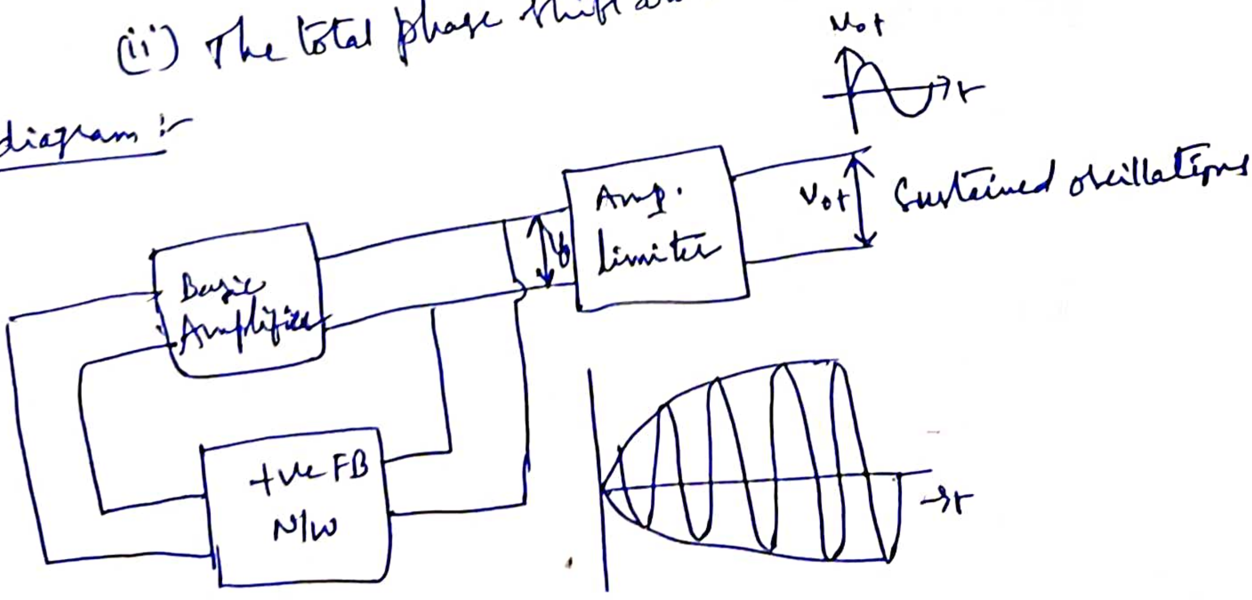
Mechanism for start of oscillations:- The oscillator circuit is set into conditions by a random variations in the base current due to noise component (or) a small variation in dc power supply. The noise components i.e, extremely small random electrical voltages or currents are always present in any conductor, transistor. Even when no external signal is applied, the ever present noise will cause some small signal at the o/p of amplifier. When the amplifier is tuned at a particular frequency f_r, the o/p signal caused by noise signal will be at f_r. If a small signal of o/p signal is feedback to the i/p with proper phase relation, then this feedback signal will be amplified. If the amplifier gain $> 1/\beta$, then the o/p increases and thereby feedback signal becomes larger. This process continues and o/p increases. As the signal level increases, the gain of the amplifier decreases and at a particular value of o/p, the gain of the amplifier is reduced exactly to $1/\beta$. Then the o/p voltage remains constant at frequency f_r, called frequency of oscillation.

The essentials conditions for maintaining oscillations are

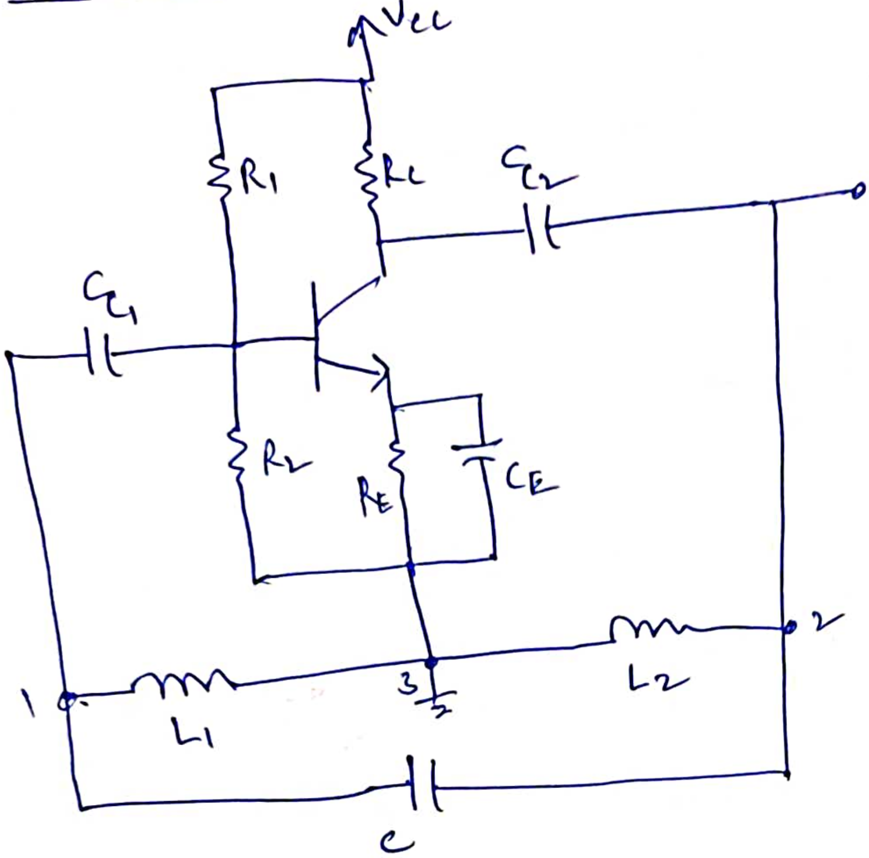
(i) $|A\beta| = 1$, loopgain $= 1$ (magnitude)

(ii) The total phase shift around closed loop is 0° or 360°.

Block diagram:-



Hartley oscillator :-



⇒ In the Hartley oscillator L_1, L_2 and C used as a tank circuit.

⇒ R_1, R_2, R_E provide the bias to the transistor.

⇒ C_E is a bypass capacitor

⇒ C_{C_1}, C_{C_2} are coupling capacitors

⇒ The FB N/w consisting of L_1, L_2 and C determines the frequency of oscillator.

Operation :- When +Vcc is switched, a transient current is produced in the tank circuit and consequently damped harmonic oscillations are set up in circuit. The oscillatory current in the tank circuit produces ac voltage across L_1 and L_2.

Terminal 1 is at a +Ve potential w.r.t 3 at any instant &

Terminal 2 is at -Ve potential w.r.t 3 at the same instant.

Thus the phase difference between 1 and 2 is always 180°.

In the CE mode, the transistor provides a phase difference of 180° b/w i/p and o/p. Therefore total phase shift is 360°.

Thus, at the frequency determined for tank circuit, the necessary condition for sustained oscillations is satisfied. If the feedback is adjusted so that the loop gain $A\beta = 1$, the circuit acts as an oscillator.

∴ Frequency of oscillation $f_r = \dfrac{1}{2\pi\sqrt{LC}}$, where $L = L_1 + L_2 + 2M$

M = Mutual Inductance b/w L_1, L_2

Condition for oscillation $h_{fe} \geq \dfrac{L_1 + M}{L_2 + M}$

Analysis :- $Z_1 = j\omega L_1 + j\omega M$, $Z_2 = j\omega L_2 + j\omega M$

$$Z_3 = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

Sub this in $h_{ie}(Z_1+Z_2+Z_3) + Z_1 Z_2 (1+h_{fe}) + Z_1 Z_3 = 0$. (general equation for oscillator)

$$j\omega h_{ie}\left[L_1+L_2+2M-\frac{1}{\omega^2 C}\right] - \omega^2 (L_1+M)\left[(L_2+M)(1+h_{fe}) - \frac{1}{\omega^2 C}\right] = 0$$

frequency of oscillation $f_r = \frac{\omega_r}{2\pi}$ can be determined by equating imaginary part to zero :-

$$L_1+L_2+2M - \frac{1}{\omega_r^2 C} = 0 \Rightarrow (L_1+L_2+2M)C = \frac{1}{\omega_r^2}$$

$$\Rightarrow \omega_r^2 = \frac{1}{(L_1+L_2+2M)C}$$

$$\Rightarrow \omega_r = \frac{1}{\sqrt{(L_1+L_2+2M)C}}$$

$$f_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi\sqrt{(L_1+L_2+2M)C}}$$

Condition for maintenance of oscillations (real part to zero) :-

$$(1+h_{fe})(L_2+M) - \frac{1}{\omega_r^2 C} = 0 \rightarrow (1)$$

Sub ω_r in the equation (1)

$$\boxed{h_{fe} = \frac{L_1+M}{L_2+M}}$$

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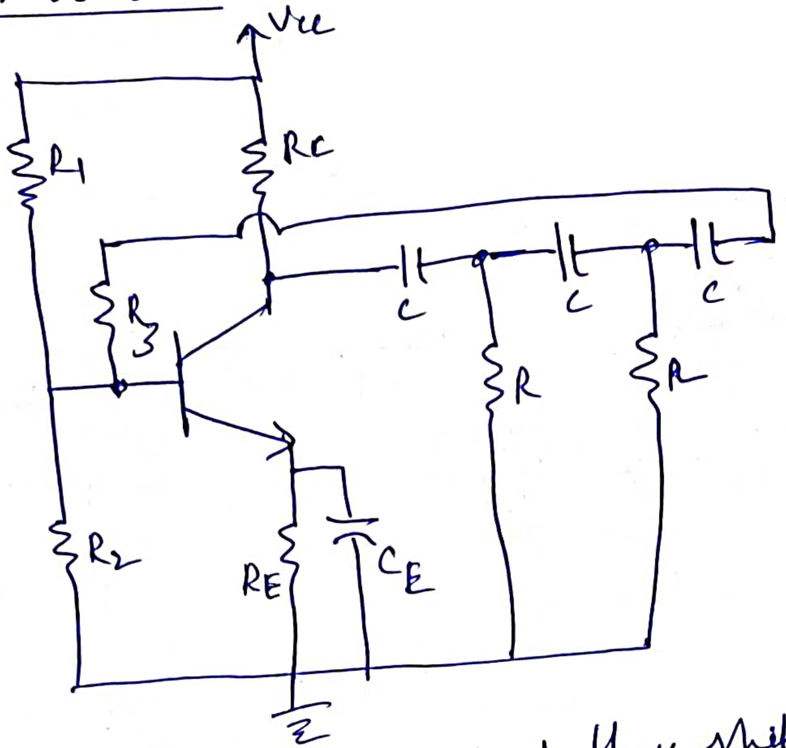
* LC circuits operate well at high frequencies.
* At low frequencies RC oscillators are used.

RC phase shift oscillator



$R_3 = R - R_i$

R_i = i/p impedance

In this oscillator the required phase shift of 180° in the FB loop from o/p to i/p is obtained by R and C components.

The circuit of RC phase shift oscillator using cascade connection of highpass filter. A Common Emitter amplifier is followed by three sections of RC phase shift N/w, the o/p of the last section being returned to i/p.

$$\text{Phase shift } \phi = \tan^{-1}\left(\frac{1}{\omega RC}\right)$$

R value is adjusted such that $\phi = 60°$ in each section.

The RC ladder N/w produces a total phase shift of 180° b/w i/p and o/p for the given frequency.

∴ At the specific frequency f_r, the total phase shift from the base of transistor around the circuit and back to base will be exactly 360° or 0°, there by satisfying Barkhausen criterion for oscillation.

The frequency of oscillation $f_r = \frac{1}{2\pi\sqrt{6}\,RC}$.

At the frequency, $|\beta| = \frac{1}{29}$.

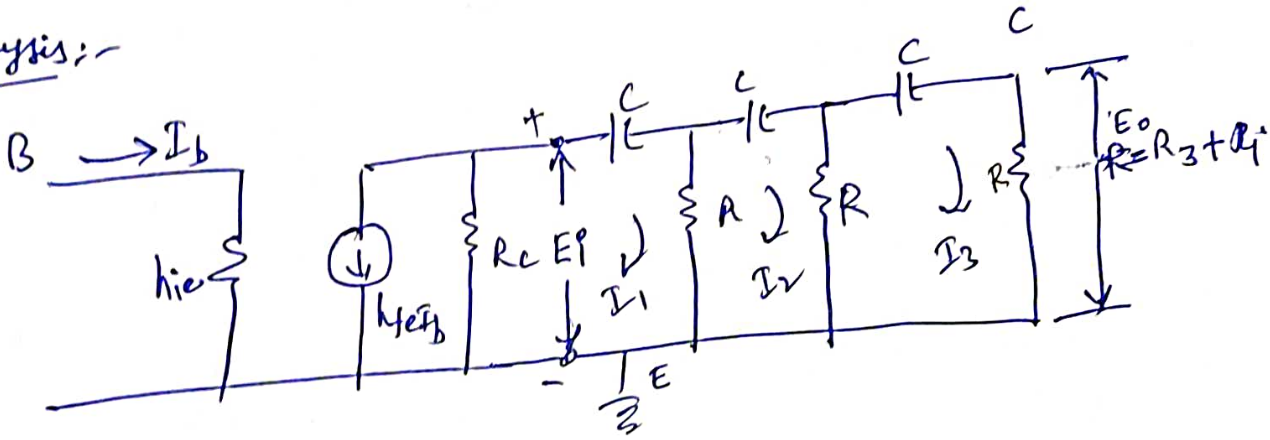
$\therefore$ It is required that the amplifier gain $|A|$ more than 29 for oscillator operation.

* It is suitable for audio frequencies only.

Drawbacks:- (1) The three capacitors or Resistors should be changed simultaneously to change the frequency of oscillation.

(2) It is difficult to control the amplitude of oscillation without affecting the frequency of oscillation.

Analysis:-



From ckt

$$I_1\left(R + \frac{1}{j\omega C}\right) - I_2 R = E_i$$

$$-I_1 R + I_2\left(2R + \frac{1}{j\omega C}\right) - I_3 R = 0$$

$$-I_2 R + \left(2R + \frac{1}{j\omega C}\right) I_3 = 0$$

solving the above simultaneous equations, we get

$$I_3 = \frac{E_i}{R}\left[\frac{1}{(1-5\alpha^2) + j\alpha(\alpha^2-6)}\right], \quad \alpha = \frac{1}{\omega RC}$$

$$E_0 = E_i \left[\frac{1}{(1-5\alpha^2)+j\alpha(\alpha^2-6)} \right]$$, where $E_0 = E_i R$

$$\beta = \frac{E_0}{E_i} = \left(\frac{1}{(1-5\alpha^2)+j\alpha(\alpha^2-6)} \right)$$

For determining the frequency of oscillation, phase shift, the imaginary part must be equal to zero.

i.e $\alpha(\alpha^2-6) = 0$ $\quad [\because \alpha \neq 0]$

$$\alpha = \frac{1}{\omega_r RC} = \sqrt{6}$$

$$\boxed{\therefore f_r = \frac{1}{2\pi\sqrt{6}\,RC}}$$

The condition for oscillation is obtained by sub $\alpha = \sqrt{6}$ in β.

Thus $\beta = -\frac{1}{29} = \frac{1}{29}\angle 180^\circ$

(or) $|\beta| = \frac{1}{29}$ $\quad$ i.e Transistor gain 'A' should not be less than 29

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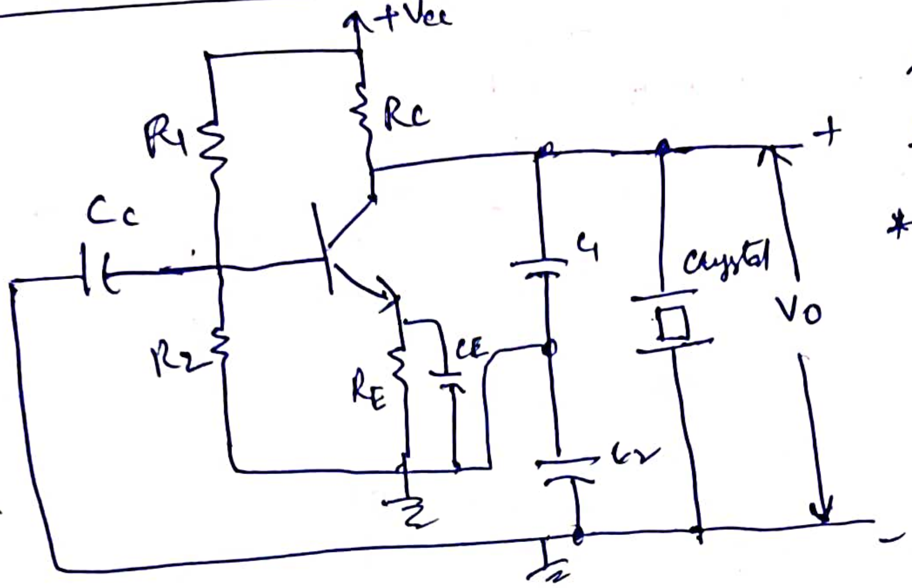
Crystal oscillators :-

Colpitts crystal oscillator



* Colpitts crystal oscillator in which the inductor is replaced by the crystal.

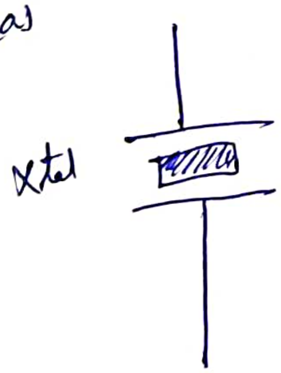
* In this type, a piezo-electric crystal (Quartz), is used as a resonant circuit replacing an LC circuit.

* A crystal is a thin slice of piezoelectric material such as quartz, rochelle salt which exhibits a property called piezo-electric effect.

* The piezo-electric effect represents the characteristics that the crystal reacts to any mechanical stress by producing an electric charge.

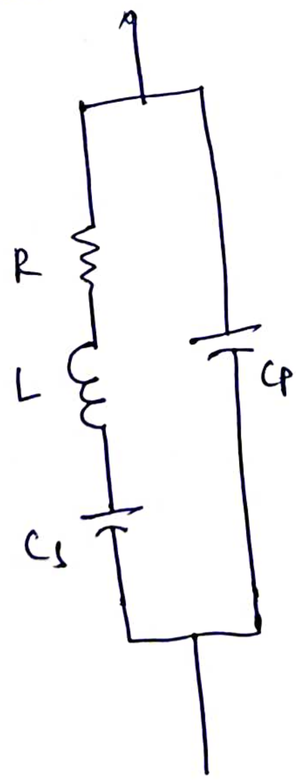
Quartz crystal construction :-

(a)



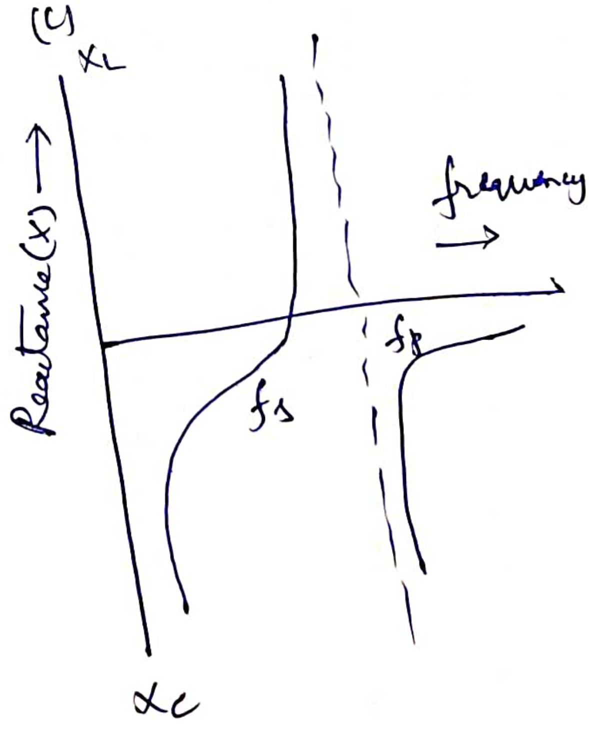
symbol

(b)



Electrical equivalent

(c)



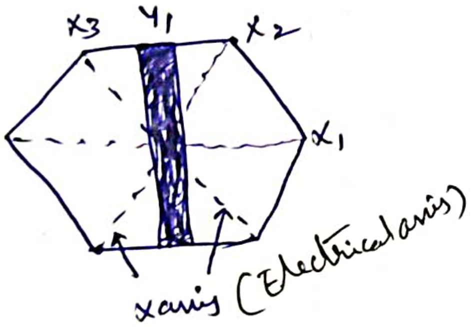
* In order to obtain high degree of frequency stability, crystal oscillators are used.

Crystal is a ground wafer of translucent quartz (or) tourmaline stone placed between two metal plates and housed into a stamp size package.

There are two methods of cutting this crystal wafer from crude quartz.

The method of cutting determines the natural resonant frequency and temperature coefficient of crystal.

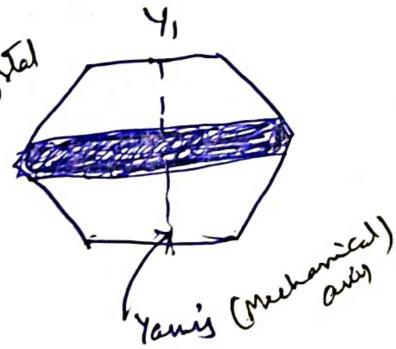
X-cut crystal



When the wafer is cut in such a way that its flat surfaces are ⊥ar to its electrical axis (X-axis)

When the wafer is cut in such a way that its flat surfaces are ⊥ar to mechanical axis (Y-axis), it is called Y-cut crystal.

Y-cut crystal



* If an ac voltage is applied, then the crystal wafer is set into vibrations. The frequency of vibration equal to resonant frequency of the crystal.

* If the frequency of applied ac voltage is equal to natural resonant frequency of ~~the~~ crystal, then the maximum amplitude of vibration will be obtained. The frequency of vibration is inversely proportional to thickness of crystal.

The frequency of vibration is $f = \frac{P}{2l}\sqrt{\frac{Y}{\rho}}$

Y = Young Modulus , ρ = density of the material , $P = 1, 2, 3 \ldots$

The crystal is suitably cut and polished to vibrate at a certain frequency and mounted b/w two metal plates.

The ratio of C_s and C_p may be several hundred or more so that series resonance frequency is very close to parallel resonant frequency.

Resonant frequencies from 0.5 to 30 MHz can be obtained.

The reactance function is given as

$$jX = \frac{1}{j\omega C_p} \cdot \frac{\omega^2 - \omega_s^2}{\omega^2 - \omega_p^2}$$

neglecting R,

$$\omega_s^2 = \frac{1}{LC_s}$$

$$\omega_p^2 = \frac{1}{L\left(\frac{1}{C_s} + \frac{1}{C_p}\right)}$$

$\because C_p \gg C_s$, $\omega_p \cong \omega_s$.

* For $\omega_s < \omega < \omega_p$, the reactance is inductive and 'ω' out of range, it is capacitive.

Advantage :- (i) Very high Q as a resonant circuit, which results in good frequency stability for oscillator.